Euler-Factor Controls for the de Bruijn–Newman Constant: Certified and Asymptotic Lower Bounds

Drafted for Edward Baker

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Abstract

For a prime p and $c > 0$ we change the Riemann ξ-function at one Euler factor. In the Newman normalization this replaces the Riemann kernel Φ by $\Phi_{c,p} = \Phi + \frac{c}{2}[\Phi(\cdot + a) + \Phi(\cdot - a)]$ with $a = \frac{1}{2}\log p$, and the time-zero transform $H_0(x)$ by $H_0(x)(1 + c\cos ax)$. The associated Dirichlet series is $\zeta(s)\big(1 + \frac{2\sqrt{p}}{c}p^{-s} + p^{1-2s}\big)$. It has every Euler factor of ζ except the one at p, lies in the extended Selberg class with conductor p^2, and its local factor at p is tempered exactly when $c \geq 1$. For $c \geq 1$ the de Bruijn–Newman constant $\Lambda_{c,p}$ satisfies $\Lambda_{c,p} \leq 0$ if and only if the Riemann hypothesis holds. For $0 < c < 1$ we prove three results. (i) The heat flow of $\Phi_{c,p}$ is that of Φ plus two copies of H_t at complex spectral shifts $x \pm 2ita$. (ii) By outward-rounded interval evaluation of one Laguerre expression, $\Lambda_{1/2,p} > 0.2$ for every prime $p \leq 41$, and $\Lambda_{1/2,p} > 0.1787854$ for $p = 43, 47$. (iii) With $t_m = 4\log(1/c)/(\log p)^2$ and $S_2 = \xi''(\frac{1}{2})/(2\xi(\frac{1}{2}))$,

$$\Lambda_{c,p} > t_m - S_2 t_m^2 - \tfrac{1}{50}\big(\tfrac{1}{2}\log p\big)^{-6} \qquad (\tfrac{1}{2} \leq c < 1,\ p \geq 29).$$

Numerically, the lowest off-line pair of zeros collides at $t_m - S_2 t_m^2 + O(t_m^3)$. Since the best refereed upper bound is $\Lambda \leq 0.2$, these kernels show that any proof of an upper bound below $\Lambda_{c,p}$ must use an input that $\Phi_{c,p}$ lacks, such as the Euler factor of ζ at p or direct information about the zeros of ξ; the controls already have non-real zeros at height $\pi/a < 10$. We describe how the existing proofs supply such inputs. No new information about the zeros of ζ is claimed.

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1 Introduction

1.1 The de Bruijn–Newman constant

Let

$$\Phi(u) = \sum_{n\geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) \exp(-\pi n^2 e^{4u}), \tag{1.1}$$

and for an even real kernel K of sufficiently rapid decay put

$$H_t[K](z) = \int_0^\infty e^{tu^2} K(u) \cos(zu)\, du, \qquad H_t = H_t[\Phi]. \tag{1.2}$$

Then $H_0(z) = \frac{1}{8}\xi(\frac{1}{2} + \frac{iz}{2})$, where $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, and $\partial_t H_t = -\partial_z^2 H_t$. De Bruijn [3] showed that real-rootedness of H_t persists as t increases and that H_t has only real zeros for $t \geq \frac{1}{2}$. Newman [7] showed that there is a constant $\Lambda \in (-\infty, \frac{1}{2}]$ such that H_t has only real zeros exactly when $t \geq \Lambda$. The Riemann hypothesis is equivalent to $\Lambda \leq 0$. Earlier lower bounds for Λ came from Lehmer pairs, pairs of unusually close zeros of ζ [2]; the witnesses of Section 5 also rest on the local behaviour of a single pair of zeros. Rodgers and Tao [12] proved $\Lambda \geq 0$; Dobner [4] gave a new proof and extended it to the extended Selberg class. On the other side, Ki, Kim and Lee [6] proved $\Lambda < \frac{1}{2}$, the Polymath project [11] proved $\Lambda \leq 0.22$, and Platt and Trudgian [9] proved $\Lambda \leq 0.2$, the best bound in the refereed literature. A computer-assisted claim $\Lambda \leq 0.1787854$ [5] has not been refereed and is not used below. The survey [8] describes the wider landscape.

1.2 Controls

An upper bound $\Lambda \leq t_0$ needs information about ζ. One way to see which information is to exhibit kernels that share many properties with Φ but have de Bruijn–Newman constant larger than t_0. Analytic shape alone gives no upper bound: $e^{-Tu^2}\Phi(u)$ is positive, even and rapidly decaying, and its constant is $\Lambda + T$, because $H_t[e^{-Tu^2}K] = H_{t-T}[K]$. This paper studies kernels that are much closer to Φ arithmetically: they differ from it at one Euler factor.

Fix a prime p, write

$$a = a_p = \tfrac{1}{2}\log p, \tag{1.3}$$

and for $c > 0$ define

$$\Phi_{c,p}(u) = \Phi(u) + \frac{c}{2}\big[\Phi(u+a) + \Phi(u-a)\big], \qquad H_t^{c,p} = H_t[\Phi_{c,p}], \tag{1.4}$$

and let $\Lambda_{c,p}$ be the de Bruijn–Newman constant of $\Phi_{c,p}$ (Definition 3.2). At $t = 0$,

$$H_0^{c,p}(x) = H_0(x)\,(1 + c\cos ax). \tag{1.5}$$

For $0 < c < 1$ the multiplier lies in $[1-c, 1+c]$ on the real line, so $H_0^{c,p}$ has exactly the real zeros of H_0. Its additional zeros lie on the lines $\operatorname{Im} x = \pm\delta$, where

$$\delta = \delta_{c,p} = \frac{\operatorname{arccosh}(1/c)}{a}. \tag{1.6}$$

We also use

$$\tau_0 = \log\frac{1}{c}, \qquad t_m = \frac{\tau_0}{a^2} = \frac{4\log(1/c)}{(\log p)^2}, \qquad S_2 = \frac{\xi''(\frac12)}{2\,\xi(\frac12)} = 0.0231049931154\ldots. \tag{1.7}$$

The quantity S_2 equals $\sum_\gamma \gamma^{-2}$, summed over the zeros $\frac12 + i\gamma$ of ξ with $\operatorname{Re}\gamma > 0$; under RH this is the sum over the positive ordinates.

1.3 Results

The first theorem collects the structural facts (Section 2 and Proposition 3.3).

Theorem 1.1 (structure)**.** *Let p be prime and $c > 0$.*

(a) *$\Phi_{c,p}$ is positive and even, and $H_t^{c,p}$ is an even entire function of order at most one for every real t.*

(b) *The Dirichlet series attached to $H_0^{c,p}$ is $F_{c,p}(s) = \zeta(s)Q_p(p^{-s})$ with $Q_p(X) = 1 + \frac{2\sqrt{p}}{c}X + pX^2$. It has positive coefficients and the Euler factor of ζ at every prime $q \neq p$. It belongs to the extended Selberg class with degree one and conductor p^2. Its local factor at p is tempered (its roots lie on $|X| = p^{-1/2}$) if and only if $c \geq 1$.*

(c) *If $c \geq 1$, then $\Lambda_{c,p} \geq 0$, and $\Lambda_{c,p} \leq 0$ if and only if RH holds.*

(d) *If $0 < c < 1$, then $0 < \Lambda_{c,p} \leq \max(1,\delta)^2/2$, and $\Lambda_{c,p} \leq \delta^2/2$ under RH.*

The heat flow of a control is determined by the heat flow of Φ.

Theorem 1.2 (exact flow)**.** *For every real t and complex z,*

$$H_t^{c,p}(z) = H_t(z) + \frac{c}{2}\,e^{ta^2}\Big[e^{-iaz}H_t(z + 2ita) + e^{iaz}H_t(z - 2ita)\Big].$$

To first order the bracket acts like the multiplier $1 + ce^{ta^2}\cos az$, whose extra zeros become real when $ce^{ta^2} = 1$, that is at $t = t_m$. The factor $e^{ta^2} = \exp(\frac{t}{4}\log^2 p)$ is the weight b_p^t with which the heat flow multiplies the p-th Dirichlet coefficient in the effective approximation of [11]. At t_m the amplified local factor reaches the tempered boundary (Remark 4.1).

A single negative value of the Laguerre expression $\mathscr{L}_0(f;x) = f'(x)^2 - f(x)f''(x)$ proves a lower bound for $\Lambda_{c,p}$ (Proposition 5.1). Interval evaluation of such witnesses gives the following.

Theorem 1.3 (certified bounds)**.** $\Lambda_{1/2,p} > 0.2$ *for every prime* $p \le 41$*, and* $\Lambda_{1/2,p} > 0.1787854$ *for* $p \in \{43, 47\}$*.*

For large p we obtain an asymptotic lower bound with explicit constants.

Theorem 1.4 (asymptotic lower bound)**.** *Let* p *be prime and* $0 < c < 1$*, and let* a, τ_0, t_m, S_2 *be as above.*

(i) *If* $\frac{1}{2} \le c < 1$ *and* $p \ge 29$*, then* $\Lambda_{c,p} > t_m - S_2 t_m^2 - \frac{1}{50}\, a^{-6}$*.*

(ii) *If* $e^{-2} \le c < 1$ *and* $p \ge 293$*, then* $\Lambda_{c,p} > t_m - S_2 t_m^2 - \frac{1}{10}\, a^{-6}$*.*

In both cases the Laguerre expression of $H_t^{c,p}$ *at* $x = \pi/a$ *is negative at the displayed time* t *whenever* $t \ge 0$*. For every fixed* $c \in (0, 1)$*,* $\Lambda_{c,p} \ge t_m - S_2 t_m^2 - O(t_m^3)$ *as* $p \to \infty$*.*

For $c = \frac{1}{2}$ the bound of Theorem 1.4(i) exceeds 0.2 for $29 \le p \le 37$ and exceeds 0.1787854 for $29 \le p \le 47$. So the analytic and the computational results confirm each other for $p \in \{29, 31, 37, 43, 47\}$; for $p = 41$ only the computation reaches 0.2. Under RH, Theorem 1.1(d) gives the upper bound $\delta^2/2 = t_m(1 + \frac{1}{3}(1 - c) + O((1 - c)^2))$, so that

$$1 - S_2 t_m - \frac{1}{50\,\tau_0 a^4} < \frac{\Lambda_{c,p}}{t_m} \le 1 + \frac{1 - c}{3} + O((1 - c)^2),$$

and both sides tend to 1 if $p \to \infty$ and then $c \to 1$ (Corollary 7.4).

Numerical tracking of the zeros (Section 8) indicates that the lowest off-line pair is the last of the tracked pairs to collide, at a time $T_0(c, p) = t_m - S_2 t_m^2 + O(t_m^3)$. We do not prove an upper bound $\Lambda_{c,p} \le T_0(c, p)$; that would require control of all other zeros.

1.4 What the controls say about proofs

Theorem 1.3 is a statement about specific entire functions. It also has a methodological reading (Section 9): any collection of hypotheses satisfied by some $\Phi_{1/2,p}$ with $p \le 41$ cannot imply $\Lambda_K \le 0.2$ for all kernels K satisfying them. The shared hypotheses include positivity, evenness and the decay of the kernel; the real zeros of H_0; bounded relative distortion of H_0 on the real line; the Euler factors and Hecke relations at every prime other than p; and membership in the extended Selberg class. The existing proofs of upper bounds avoid this obstruction by using inputs that the controls do not have, namely numerical verification of RH up to a large height and effective approximations built from the Dirichlet coefficients and small Euler factors of ζ. The controls make precise that these direct inputs, which the controls fail already at height $\pi/a \le 9.1$, cannot be replaced by structural facts shared with $\Phi_{c,p}$.

1.5 Organization

Section 2 gives the arithmetic of the controls. Section 3 recalls the de Bruijn–Newman theory for general kernels and proves Theorem 1.1. Section 4 proves Theorem 1.2. Section 5 treats Laguerre witnesses. Section 6 proves Theorem 1.3, and Section 7 proves Theorem 1.4. Section 8 reports the numerical collision times, Section 9 discusses consequences for proof strategies, and Section 10 lists open problems.

2 The controls

Lemma 2.1. *Let p be prime and $c > 0$.*

(a) *$\Phi_{c,p}$ is positive and even on $\mathbb{R}$, and $\Phi_{c,p}(u) = O(\exp(9|u| - \pi e^{4|u|-4a}))$.*

(b) *For every real t, $H_t^{c,p}$ is an even entire function, real on the real axis, of order at most one.*

(c) *$H_0^{c,p}(x) = H_0(x)(1 + c\cos ax)$.*

Proof. The function Φ is even, by the functional equation of the theta function, and positive on $\mathbb{R}$ (Pólya [10]; see also [1]). Hence $\Phi_{c,p}$ is positive, and it is even because the bracket in (1.4) is. For $u \geq 0$ the series (1.1) gives $0 < \Phi(u) \leq Ce^{9u}\exp(-\pi e^{4u})$, and by evenness this holds with $|u|$ in place of u. Since $|u \pm a| \geq |u| - a$, part (a) follows.

For complex z with $|\operatorname{Im} z| = y$, $|H_t^{c,p}(z)| \leq \int_0^\infty e^{tu^2}\Phi_{c,p}(u)\cosh(yu)\,du$. By (a) the integrand is at most $C\exp(tu^2 + (9+y)u - \pi e^{4u-4a})$, and the maximum over u of the exponent is $O(y\log y)$ as $y \to \infty$. Hence $|H_t^{c,p}(z)| \leq \exp(O(|z|\log|z|))$, which gives order at most one. The function is even and real on $\mathbb{R}$ because the kernel is even and real.

For (c), extend Φ evenly and write $H_0[K](x) = \frac{1}{2}\int_{\mathbb{R}} K(u)e^{ixu}\,du$ for even K. Then $\frac{1}{2}\int_{\mathbb{R}}[\Phi(u+a) + \Phi(u-a)]e^{ixu}\,du = (e^{-iax} + e^{iax})H_0(x) = 2\cos(ax)\,H_0(x)$. □

Proposition 2.2 (arithmetic)**.** *Let $s = \frac{1}{2} + \frac{ix}{2}$ and $Q_p(X) = 1 + \frac{2\sqrt{p}}{c}X + pX^2$.*

(a) *$1 + c\cos ax = \frac{c}{2}\,p^{s-1/2}Q_p(p^{-s})$, so $H_0^{c,p}(x) = \frac{c}{16}\,p^{s-1/2}\,\xi(s)\,Q_p(p^{-s})$.*

(b) *$F_{c,p}(s) = \zeta(s)Q_p(p^{-s}) = \sum_{n\geq 1} a_n n^{-s}$ with $a_n = 1 + \frac{2\sqrt{p}}{c}\mathbf{1}_{p|n} + p\,\mathbf{1}_{p^2|n} > 0$, and*

$$F_{c,p}(s) = \frac{Q_p(p^{-s})}{1-p^{-s}}\prod_{q\neq p}\frac{1}{1-q^{-s}} \qquad (\operatorname{Re} s > 1).$$

(c) *The completed function $\Lambda^{\mathrm{L}}_{c,p}(s) = (p/\sqrt{\pi})^s\,\Gamma(s/2)F_{c,p}(s)$ satisfies $\Lambda^{\mathrm{L}}_{c,p}(s) = \Lambda^{\mathrm{L}}_{c,p}(1-s)$. Thus $F_{c,p}$ belongs to the extended Selberg class $\mathcal{S}^\sharp$, with degree one and conductor p^2.*

(d) *The roots $X_\pm$ of Q_p satisfy $X_+X_- = 1/p$. If $c \geq 1$ they lie on $|X| = p^{-1/2}$, and they coincide when $c = 1$. If $0 < c < 1$ they are real and negative, with $|X_\pm| = p^{-1/2}e^{\mp\operatorname{arccosh}(1/c)}$. Consequently the zeros of $Q_p(p^{-s})$ lie on $\operatorname{Re} s = \frac{1}{2}$ if $c \geq 1$. If $c < 1$ they lie on $\operatorname{Re} s = \frac{1}{2} \pm \operatorname{arccosh}(1/c)/\log p$, which in the variable x are the lines $\operatorname{Im} x = \pm\delta$.*

Proof. (a) Here $p^{s-1/2} = e^{iax}$ and $p^{-s} = p^{-1/2}e^{-iax}$, so $\frac{c}{2}p^{s-1/2}Q_p(p^{-s}) = \frac{c}{2}e^{iax} + 1 + \frac{c}{2}e^{-iax}$. Combine with (1.5) and $H_0 = \xi/8$. (b) is the expansion of $\zeta(s)Q_p(p^{-s})$. (c) By (a), $\Lambda^{\mathrm{L}}_{c,p}(s) = \frac{2\sqrt{p}}{s(s-1)}\cdot\frac{2}{c}\,(1 + c\cos ax)\,\xi(s)$. Each factor is invariant under $s \mapsto 1-s$, that is $x \mapsto -x$. The function $F_{c,p}$ is a Dirichlet series absolutely convergent for $\operatorname{Re} s > 1$, $(s-1)F_{c,p}$ is entire of finite order, and the functional equation has the shape required in [4]. The conductor is $(2\pi)\cdot(p/\sqrt{\pi})^2\cdot\frac{1}{2} = p^2$. (d) The discriminant of $pX^2 + \frac{2\sqrt{p}}{c}X + 1$ is $4p(c^{-2}-1)$. If $c \geq 1$ the roots are conjugate, or equal when $c = 1$, and $|X_\pm|^2 = X_+X_- = 1/p$. If $c < 1$, then $X_\pm = -p^{-1/2}(c^{-1} \mp \sqrt{c^{-2}-1}) = -p^{-1/2}e^{\mp\operatorname{arccosh}(1/c)}$. Finally $p^{-s} = X$ forces $\operatorname{Re} s = -\log|X|/\log p$, and $\operatorname{Im} x = -2(\operatorname{Re} s - \frac{1}{2})$. □

Remark 2.3 (Hecke interpretation). For the normalized Hecke operator $T_pf(z) = p^{-1/2}(f(pz) + \sum_{j=0}^{p-1} f((z+j)/p))$, the Eisenstein series $E(z, \frac{1}{2} + i\nu)$ has eigenvalue $\lambda_p(\nu) = p^{i\nu} + p^{-i\nu} = 2\cos(\nu\log p)$. With $\nu = x/2$,

$$1 + c\cos ax = 1 + \tfrac{c}{2}\,\lambda_p(x/2).$$

Thus $H_0^{c,p}$ is the spectral density of Φ multiplied by the eigenvalue of the Hecke element $1+\frac{c}{2}T_p$. A multiplier that depends only on λ_p leaves the local data at every prime $q \neq p$ unchanged; this is the spectral form of Proposition 2.2(b).

Remark 2.4 (the Selberg class). $F_{c,p}$ has an Euler product, but its local factor at p, $Q_p(p^{-s})/(1-p^{-s})$, has zeros. The coefficients of $\log F_{c,p}$ at p^k are $-(X_+^{-k}+X_-^{-k})/k$ plus the contribution of $(1-p^{-s})^{-1}$, and they have size at least a constant times $p^{k/2}/k$ for infinitely many k. So $F_{c,p}$ violates the Euler-product axiom of the Selberg class for every $c>0$, while it lies in $\mathcal{S}^\sharp$, which imposes no such axiom.

3 De Bruijn–Newman constants of the controls

Lemma 3.1. *Let K be a real even kernel with $K(u)=O(\exp(-|u|^{2+\eta}))$ for some $\eta>0$.*

(a) (de Bruijn [3, Theorem 13]; see also [11, Theorem 3.2]) *If all zeros of $H_{t_0}[K]$ lie in $|\operatorname{Im} z|\le\Delta$, then for $t\ge t_0$ all zeros of $H_t[K]$ lie in $|\operatorname{Im} z|\le\sqrt{\max(\Delta^2-2(t-t_0),0)}$. In particular, real-rootedness persists as t increases.*

(b) *The set Z_K of $t\in\mathbb{R}$ for which $H_t[K]$ has only real zeros is closed.*

(c) *If $H_t[K]$ is of order less than two and has only real zeros, then $\mathscr{L}_0(H_t[K];x)\ge 0$ for every real x.*

Proof. (b) The map $t\mapsto H_t[K]$ is continuous into the entire functions with the topology of locally uniform convergence, and $H_t[K]\not\equiv 0$. By Hurwitz's theorem, a locally uniform limit of entire functions with only real zeros has only real zeros. (c) By Hadamard's theorem, an entire function f of order less than two, real on $\mathbb{R}$, with only real zeros x_k, has the form $f(z)=Cz^m e^{\beta z}\prod_k(1-z/x_k)e^{z/x_k}$ with real β. For real x that is not a zero, $-(\log f)''(x)=mx^{-2}+\sum_k(x-x_k)^{-2}\ge 0$, so $\mathscr{L}_0(f;x)=-f(x)^2(\log f)''(x)\ge 0$. At a zero, $\mathscr{L}_0(f;x)=f'(x)^2\ge 0$. □

Definition 3.2. For K as in Lemma 3.1 put $\Lambda_K=\inf Z_K\in[-\infty,\infty]$. By Lemma 3.1, $Z_K=[\Lambda_K,\infty)$ whenever Λ_K is finite. We write $\Lambda=\Lambda_\Phi$ and $\Lambda_{c,p}=\Lambda_{\Phi_{c,p}}$.

Lemma 2.1(a) shows that $\Phi_{c,p}$ satisfies the hypothesis of Lemma 3.1.

Proposition 3.3. *Let p be prime.*

(a) *If $0<c<1$, then $0<\Lambda_{c,p}\le\max(1,\delta)^2/2$, and under RH $\Lambda_{c,p}\le\delta^2/2$.*

(b) *If $c\ge 1$, then $\Lambda_{c,p}\ge 0$, and $\Lambda_{c,p}\le 0$ holds if and only if RH holds. Under RH, $\Lambda_{c,p}=0$.*

(c) *For every $c>0$, $\Lambda_{c,p}\ge 0$.*

Proof. By Lemma 2.1(c), the zeros of $H_0^{c,p}$ are those of H_0, which lie in $|\operatorname{Im} x|<1$ because the nontrivial zeros of ζ lie in $0<\operatorname{Re} s<1$, together with the zeros of $1+c\cos ax$. (a) For $c<1$ the latter lie on $|\operatorname{Im} x|=\delta$ (Proposition 2.2(d)). Lemma 3.1(a) with $\Delta=\max(1,\delta)$, or $\Delta=\delta$ under RH, gives the upper bounds. Since $H_0^{c,p}$ has non-real zeros, $0\notin Z_{\Phi_{c,p}}$, hence $\Lambda_{c,p}>0$. (b) For $c\ge 1$ the zeros of $1+c\cos ax$ are real. Hence $H_0^{c,p}$ has only real zeros if and only if H_0 does, that is, if and only if RH holds. By Definition 3.2 this is equivalent to $\Lambda_{c,p}\le 0$. (c) By Proposition 2.2(c), $F_{c,p}\in\mathcal{S}^\sharp$. Its completed function in the sense of [4, Section 2] is a constant multiple of $H_0^{c,p}((s-\frac12)\cdot\frac{2}{i})$, and Dobner's heat deformation agrees with $H_t^{c,p}$ up to the same constant. Dobner's theorem [4, Theorem 2] gives $\Lambda_{c,p}\ge 0$. Together with (b) this proves the remaining claims. □

Theorem 1.1 is Lemma 2.1, Proposition 2.2 and Proposition 3.3.

4 The exact flow

Proof of Theorem 1.2. Extend Φ evenly and write $N_t[K](z) = \frac{1}{2}\int_{\mathbb{R}} e^{tu^2}K(u)e^{izu}\,du$, so that $N_t[\Phi] = H_t$ and $H_t^{c,p} = H_t + \frac{c}{2}(N_t[\Phi(\cdot+a)] + N_t[\Phi(\cdot-a)])$. Substituting $v = u \pm a$,

$$N_t[\Phi(\cdot \pm a)](z) = \tfrac{1}{2}\int_{\mathbb{R}} e^{t(v\mp a)^2}\Phi(v)e^{iz(v\mp a)}\,dv = e^{ta^2}e^{\mp iaz}\,\tfrac{1}{2}\int_{\mathbb{R}} e^{tv^2}\Phi(v)e^{i(z\pm 2ita)v}\,dv,$$

which is $e^{ta^2}e^{\mp iaz}H_t(z \pm 2ita)$. All integrals converge absolutely by Lemma 2.1(a). □

For real x, $H_t(x - 2ita) = \overline{H_t(x + 2ita)}$, so

$$H_t^{c,p}(x) = H_t(x) + C\ \mathrm{Re}\big[e^{-iax}H_t(x + 2ita)\big], \qquad C = ce^{ta^2}. \tag{4.1}$$

Remark 4.1 (the model multiplier). Replacing $H_t(x \pm 2ita)$ by $H_t(x)$ in Theorem 1.2 gives $H_t(x)(1 + C\cos ax)$. With $C = ce^{ta^2}$, the extra zeros of the multiplier are non-real for $C < 1$, double for $C = 1$ and real for $C > 1$. They become real at $t = t_m = \tau_0/a^2$. By Proposition 2.2(d), the condition $C \geq 1$ says that the local factor Q_p with c replaced by C is tempered. The amplification $e^{ta^2} = \exp(\frac{t}{4}\log^2 p)$ is the weight $b_p^t = \exp(\frac{t}{4}\log^2 p)$ that the heat flow attaches to the p-th Dirichlet coefficient in the effective approximation of [11]. To first order, the control collides when that weight has amplified its local perturbation to the tempered boundary. The shifts $\pm 2ita$ produce the correction $-S_2 t_m^2$ of Theorem 1.4.

5 Laguerre witnesses

Proposition 5.1 (witness criterion)**.** *If $\mathscr{L}_0(H_t^{c,p}; x_0) < 0$ for some real x_0, then $H_t^{c,p}$ has a non-real zero and $\Lambda_{c,p} > t$.*

Proof. If $t \geq \Lambda_{c,p}$, then $t \in Z_{\Phi_{c,p}}$ by Definition 3.2, so $H_t^{c,p}$ has only real zeros. It has order at most one by Lemma 2.1(b), so Lemma 3.1(c) gives $\mathscr{L}_0(H_t^{c,p}; x) \geq 0$ for all real x. □

Near an isolated pair of non-real zeros the Laguerre expression has a simple local form.

Proposition 5.2. *Suppose that $H(x) = ((x - X)^2 + y^2)\,g(x)$ near a real point X, with g smooth and real and $y > 0$. Then*

$$\mathscr{L}_0(H; X) = -2y^2 g(X)^2 + y^4\mathscr{L}_0(g; X), \qquad \frac{\mathscr{L}_0(H; X)}{H(X)^2} = -\frac{2}{y^2} + \frac{\mathscr{L}_0(g; X)}{g(X)^2}.$$

Proof. With $q(x) = (x - X)^2 + y^2$ one has $q(X) = y^2$, $q'(X) = 0$ and $q'' = 2$. Then $H(X) = y^2 g$, $H'(X) = y^2 g'$ and $H''(X) = 2g + y^2 g''$, evaluated at X. So $H'^2 - HH'' = y^4(g'^2 - gg'') - 2y^2 g^2$. □

Remark 5.3. As a pair approaches an ordinary collision, with g, g', g'' and $1/g$ locally bounded, a witness exists at the real part of the pair. The normalized value tends to $-\infty$ like $-2/y^2$, but the raw value $\mathscr{L}_0(H; X) = -2y^2 g(X)^2 + O(y^4)$ tends to 0. A certification must therefore resolve a raw negative number, not merely a large normalized ratio. In the computations of Section 6 the raw values are at least $5 \cdot 10^{-6}$ in absolute value, and their enclosures are narrower than 10^{-51}.

Far from a collision, witnesses need not exist, since the other zeros contribute positive terms $(x - x_k)^{-2}$ to $\mathscr{L}_0/H^2$. For example, for $(p, c) = (2, 0.05)$ at $t = 0$ the off-line zero lies at about $9.065 + 10.642i$. A numerical scan nevertheless found $\mathscr{L}_0(H_0^{c,p}) > 0$ at 1200 points of $(0, 60]$.

6 Certified lower bounds

6.1 Statement

Theorem 1.3 follows from Proposition 5.1 and the following computation.

Proposition 6.1. *For $c = \frac{1}{2}$ and each row of Table 1, the enclosure of $\mathscr{L}_0(H_t^{1/2,p}; x_0)$ is contained in $(-\infty, 0)$. The table lists the upper endpoint, rounded upward.*

p	t	x_0	$H_t^{1/2,p}(x_0)$	upper bound for $\mathscr{L}_0$	$\mathscr{L}_0/H^2$
2	0.2	9.0194	$1.8797 \cdot 10^{-2}$	$-4.046 \cdot 10^{-5}$	-0.1145
3	0.2	5.6906	$2.4165 \cdot 10^{-2}$	$-1.945 \cdot 10^{-4}$	-0.3332
5	0.2	3.8844	$2.4561 \cdot 10^{-2}$	$-5.144 \cdot 10^{-4}$	-0.8529
7	0.2	3.2128	$2.3184 \cdot 10^{-2}$	$-7.795 \cdot 10^{-4}$	-1.450
11	0.2	2.6072	$1.9911 \cdot 10^{-2}$	$-1.149 \cdot 10^{-3}$	-2.900
13	0.2	2.4374	$1.8303 \cdot 10^{-2}$	$-1.268 \cdot 10^{-3}$	-3.786
17	0.2	2.2066	$1.5239 \cdot 10^{-2}$	$-1.395 \cdot 10^{-3}$	-6.011
19	0.2	2.1232	$1.3781 \cdot 10^{-2}$	$-1.411 \cdot 10^{-3}$	-7.433
23	0.2	1.9939	$1.0997 \cdot 10^{-2}$	$-1.359 \cdot 10^{-3}$	-11.24
29	0.2	1.8566	$7.0966 \cdot 10^{-3}$	$-1.094 \cdot 10^{-3}$	-21.74
31	0.2	1.8206	$5.8577 \cdot 10^{-3}$	$-9.624 \cdot 10^{-4}$	-28.05
37	0.2	1.7314	$2.2927 \cdot 10^{-3}$	$-4.443 \cdot 10^{-4}$	-84.53
41	0.2	1.6835	$2.4913 \cdot 10^{-5}$	$-5.291 \cdot 10^{-6}$	-8525
43	0.1787854	1.6622	$3.4657 \cdot 10^{-3}$	$-7.142 \cdot 10^{-4}$	-59.47
47	0.1787854	1.6238	$1.6833 \cdot 10^{-3}$	$-3.750 \cdot 10^{-4}$	-132.4

Table 1: Certified Laguerre witnesses for $c = \frac{1}{2}$. The values of H and $\mathscr{L}_0/H^2$ are rounded midpoints of their enclosures. The enclosures of H are narrower than $2 \cdot 10^{-50}$, those of $\mathscr{L}_0$ narrower than 10^{-51}, and those of $\mathscr{L}_0/H^2$ narrower than 10^{-41}. Each x_0 is a point near the real part of the lowest off-line pair.

For $p = 41$ the pair is close to its collision. Tracking gives $T_0(\frac{1}{2}, 41) \approx 0.2001167$ (Section 8). The normalized witness is large, but the raw value is only $-5.3 \cdot 10^{-6}$, as Remark 5.3 predicts. For $p \geq 43$ the lowest pair has already collided by $t = 0.2$, so the bound is certified at $t = 0.1787854$ instead.

6.2 Method

All integrands below are even, so $\int_0^\infty F = \frac{1}{2}\int_{\mathbb{R}} F$. We use the trapezoid rule on the real line.

Lemma 6.2 (Trefethen–Weideman [13], Theorem 5.1)**.** *Let F be analytic in the strip $|\operatorname{Im} u| < b$, with $F(u) \to 0$ uniformly as $|u| \to \infty$ in the strip, and $\int_{\mathbb{R}} |F(u+iv)|\,du \leq M$ for $|v| < b$. Then for every $h > 0$,*

$$\Big|\int_{\mathbb{R}} F(u)\,du - h\sum_{k\in\mathbb{Z}} F(kh)\Big| \leq \frac{2M}{e^{2\pi b/h} - 1}.$$

Lemma 6.3 (strip majorant)**.** *The function Φ extends analytically to the strip $|\operatorname{Im} w| < \pi/8$, where it is even. For $0 \leq b < \pi/8$, $\kappa = \cos 4b$ and $|\operatorname{Im} w| \leq b$,*

$$|\Phi(w)| \leq \widetilde{\Phi}_\kappa(|\operatorname{Re} w|), \qquad \widetilde{\Phi}_\kappa(r) = \sum_{n\geq 1} (2\pi^2 n^4 e^{9r} + 3\pi n^2 e^{5r}) \exp(-\pi\kappa n^2 e^{4r}).$$

Proof. For $|\operatorname{Im} w| < \pi/8$ we have $\operatorname{Re} e^{4w} = e^{4\operatorname{Re} w}\cos(4\operatorname{Im} w) > 0$, and

$$|\exp(-\pi n^2 e^{4w})| = \exp(-\pi n^2 e^{4\operatorname{Re} w}\cos(4\operatorname{Im} w)).$$

Hence the series (1.1) converges locally uniformly on the strip and defines an analytic function there, which equals Φ on $\mathbb{R}$ and is therefore even. For $\operatorname{Re} w \geq 0$ the displayed identity gives the bound termwise, using $\cos(4\operatorname{Im} w) \geq \cos 4b = \kappa$. For $\operatorname{Re} w < 0$ apply the bound to $-w$. □

Lemma 6.4 (tails)**.** *Let $\kappa \in [\frac{1}{3}, 1]$ and $r \geq 2$. Then $\widetilde{\Phi}_\kappa(r) \leq 2(2\pi^2 + 3\pi)\exp(9r - \pi\kappa e^{4r})$, and the right side decreases in r. For $N \geq 8$ and $r \geq 0$,*

$$\sum_{n>N} (2\pi^2 n^4 e^{9r} + 3\pi n^2 e^{5r}) e^{-\pi\kappa n^2 e^{4r}} \leq \frac{(2\pi^2 + 3\pi)9^4 e^{9r} e^{-81\pi\kappa e^{4r}}}{1 - (10/9)^4 e^{-19\pi\kappa e^{4r}}}.$$

Proof. Use $3\pi n^2 e^{5r} \leq 3\pi n^4 e^{9r}$ for $r \geq 0$. The ratio of consecutive terms of $n^4 e^{-\pi\kappa n^2 E}$ is at most $(10/9)^4 e^{-19\pi\kappa E}$ for $n \geq 9$. For the first bound, the factor $\sum_n n^4 e^{-\pi\kappa(n^2-1)e^{4r}}$ is at most 2 when $r \geq 2$, because $\pi\kappa e^8 > 3000$; and $9 - 4\pi\kappa e^{4r} < 0$ there. □

The computation. Fix c, p, t, x_0. The functions

$$F_0(u) = e^{tu^2}\Phi_{c,p}(u)\cos x_0 u, \quad F_1(u) = -ue^{tu^2}\Phi_{c,p}(u)\sin x_0 u, \quad F_2(u) = -u^2 e^{tu^2}\Phi_{c,p}(u)\cos x_0 u$$

are even, and $\int_0^\infty F_j$ equals H, H' and H'' at x_0. Take $b = 3/10$, so that $\kappa = \cos(6/5) > 0.36$, and $h = 1/64$. For $|\operatorname{Im} u| \leq b$, Lemma 6.3 gives

$$|F_j(u+iv)| \leq \cosh(x_0 b)\,(u^2+b^2)^{j/2} e^{tu^2}\Big[\widetilde{\Phi}_\kappa(|u|) + \tfrac{c}{2}\big(\widetilde{\Phi}_\kappa(|u+a|) + \widetilde{\Phi}_\kappa(|u-a|)\big)\Big].$$

We bound M_j by an interval Riemann upper sum of this majorant over $[0, U]$, with $U \geq a + 2$ and cells of width $1/16$, doubled by evenness. The contribution of $|u| > U$ is bounded with Lemma 6.4, using $e^{4s} \geq 1 + 4s + 8s^2$; it is below 10^{-1400}. The discretization error of Lemma 6.2 for the half-line integral is then at most $M_j/(e^{2\pi b/h} - 1)$, which is below 10^{-50} in every case. The trapezoid sum is truncated at U, and the omitted terms are bounded by the same tail estimate. On the grid $u = k/64$ the shifted thetas use the exact relations $e^{4(u\pm a)} = p^{\pm 2}e^{4u}$, $e^{5(u\pm a)} = p^{\pm 5/2}e^{5u}$ and $e^{9(u\pm a)} = p^{\pm 9/2}e^{9u}$. For $u < a$, the evenness of Φ is used. Eight terms of each theta series are summed explicitly, and the rest is bounded by Lemma 6.4.

Arithmetic. Every quantity is an interval with 60-digit decimal endpoints, rounded outward at each operation. Python's `decimal` module computes exp and log correctly rounded, and we widen the result by one unit in the last place. Negation is exact. Any operation performed outside the two directed-rounding contexts must be exact: the default context is set to raise an exception on any inexact or rounded result. Square roots of integers are verified by squaring. We compute π by Machin's formula with alternating-series remainders, and sin and cos by Taylor series after reduction by 2π, with Lagrange remainders. As consistency checks, every case was recomputed with $h = 1/32$, and the two enclosures of each of H, H', H'' overlap; the enclosure of π is compared with 70 known decimals; and the enclosure of $m_0(0) = \xi(\frac{1}{2})/8$ of Lemma 7.1 is compared with an independent value of $\xi(\frac{1}{2})$. The program uses only the Python standard library and runs in about 45 seconds; its location is given in the acknowledgements.

7 An asymptotic lower bound

7.1 Moments

For $t \ge 0$ and $r \ge 0$ put

$$m_k(t) = \int_0^\infty e^{tu^2}\Phi(u)u^k\,du, \qquad \mu_4(t,r) = \int_0^\infty e^{tu^2}\Phi(u)u^4\cosh(ru)\,du, \tag{7.1}$$

and $k_t = m_2(t)/m_0(t)$. Since $\Phi > 0$, all of these are positive and increase in t and r. Moreover $H_t(0) = m_0(t)$ and $H_t''(0) = -m_2(t)$. Since $H_0(z) = \xi(\frac12 + \frac{iz}{2})/8$, we get $k_0 = \xi''(\frac12)/(4\xi(\frac12))$, that is,

$$S_2 = 2k_0. \tag{7.2}$$

Also $\frac{d}{dt}k_t = (m_4m_0 - m_2^2)/m_0^2 \in [0, m_4/m_0]$ by the Cauchy–Schwarz inequality.

Lemma 7.1. *The following enclosures hold.*

quantity	*value*	*enclosure width*
$m_0(0) = \xi(\frac12)/8$	$0.06214009727353926373\ldots$	$< 2\cdot 10^{-51}$
$m_2(0)$	$7.17873259848294922976\ldots\cdot 10^{-4}$	$< 10^{-51}$
$S_2 = 2m_2(0)/m_0(0)$	$0.02310499311541897078\ldots$	$< 10^{-50}$
$m_2(\frac14)$	$7.23696857055556974027\ldots\cdot 10^{-4}$	$< 10^{-51}$
$m_4(\frac14)$	$2.34423514799060488665\ldots\cdot 10^{-5}$	$< 10^{-52}$
$\mu_4(\frac14, 3)$	$2.90765558946641021959\ldots\cdot 10^{-5}$	$< 10^{-51}$

Consequently, for $0 \le t \le \frac14$ *and* $0 \le r \le 3$,

$$k_t \le \bar k := 0.0116463, \qquad k_t - k_0 \le \bar d\,t, \quad \bar d := 0.000378, \qquad \frac{\mu_4(t,r)}{m_0(t)} \le \bar\mu := 0.00046792. \tag{7.3}$$

Proof. The enclosures are computed by the method of Section 6 with Φ in place of $\Phi_{c,p}$. The weights u^k and $\cosh(ru)$ are bounded on the strip by $(u^2+b^2)^{k/2}$ and $e^{r|u|}$. The constants follow from monotonicity: $m_0(t) \ge m_0(0)$, $m_2(t) \le m_2(\frac14)$, $m_4(t) \le m_4(\frac14)$ and $\mu_4(t,r) \le \mu_4(\frac14, 3)$. The program checks that the rounded values in (7.3) are upper bounds for the certified ratios, and it uses these rounded values below. □

7.2 Local expansion

Let $A(z) = H_t(z)/H_t(0)$. Expanding the cosine,

$$A(z) = 1 - \tfrac12 k_t z^2 + \rho(z), \qquad \rho(z) = \frac{1}{m_0(t)}\int_0^\infty e^{tu^2}\Phi(u)\bigl[\cos zu - 1 + \tfrac12(zu)^2\bigr]\,du. \tag{7.4}$$

Lemma 7.2. *For complex* z *with* $|z| \le R$,

$$|\rho(z)| \le \frac{|z|^4}{24}\,\mu, \qquad |\rho'(z)| \le \frac{|z|^3}{6}\,\mu, \qquad |\rho''(z)| \le \frac{|z|^2}{2}\,\mu, \qquad \mu = \frac{\mu_4(t,R)}{m_0(t)}.$$

Proof. With $y = |zu|$: $|\cos zu - 1 + \frac12(zu)^2| \le \cosh y - 1 - \frac12 y^2 \le \frac{1}{24}y^4\cosh y$. Next, $|-u\sin zu + zu^2| \le |u|(\sinh y - y) \le \frac16|u|y^3\cosh y$. Finally, $|u^2(1-\cos zu)| \le u^2(\cosh y - 1) \le \frac12 u^2y^2\cosh y$. Each comparison is termwise in the power series. Integrate against the positive weight $e^{tu^2}\Phi(u)$ and use $\cosh y \le \cosh(Ru)$. □

7.3 The witness at $x_0 = \pi/a$

Fix $0 < c < 1$, a prime p, and $0 \le t < t_m$. Put $\tau = ta^2$, $C = ce^{\tau} = e^{\tau-\tau_0} < 1$ and $\varepsilon = a^{-2}$. For real θ set $x = (\pi+\theta)/a$, so that $e^{-iax} = -e^{-i\theta}$. By (4.1),

$$F(\theta) := \frac{H_t^{c,p}((\pi+\theta)/a)}{H_t(0)} = A\Big(\frac{\pi+\theta}{a}\Big) - C\ \mathrm{Re}\Big[e^{-i\theta}A\Big(\frac{\pi+\theta+2i\tau}{a}\Big)\Big]. \tag{7.5}$$

Since $\mathscr{L}_0(H_t^{c,p};\pi/a) = a^2 m_0(t)^2(F'(0)^2 - F(0)F''(0))$, a witness at π/a means $F(0)F''(0) > F'(0)^2$. Write $x_0 = \pi/a$ and $w_0 = (\pi+2i\tau)/a$. Differentiating (7.5),

$$\begin{aligned}
F(0) &= A(x_0) - C\,\mathrm{Re}\,A(w_0),\\
F'(0) &= A'(x_0)/a - C\big[\mathrm{Im}\,A(w_0) + \mathrm{Re}\,A'(w_0)/a\big],\\
F''(0) &= A''(x_0)/a^2 + C\,\mathrm{Re}\,A(w_0) - 2C\,\mathrm{Im}\,A'(w_0)/a - C\,\mathrm{Re}\,A''(w_0)/a^2.
\end{aligned}$$

Inserting (7.4), with $k = k_t$,

$$F(0) = (1-C)(1-\tfrac12 k\pi^2\varepsilon) - 2Ck\tau^2\varepsilon + \rho(x_0) - C\,\mathrm{Re}\,\rho(w_0), \tag{7.6}$$

$$F'(0) = -k\pi\varepsilon(1-C-2C\tau) + \rho'(x_0)/a - C\,\mathrm{Im}\,\rho(w_0) - C\,\mathrm{Re}\,\rho'(w_0)/a, \tag{7.7}$$

$$F''(0) = C - k\varepsilon(1-C) - \tfrac12 Ck\pi^2\varepsilon + 2Ck\tau^2\varepsilon + 4Ck\tau\varepsilon + R_2, \tag{7.8}$$

where $R_2 = \rho''(x_0)\varepsilon + C\,\mathrm{Re}\,\rho(w_0) - 2C\,\mathrm{Im}\,\rho'(w_0)/a - C\,\mathrm{Re}\,\rho''(w_0)\varepsilon$.

Lemma 7.3. *Suppose $\tau_0 \le \bar\tau_0$, $\varepsilon \le \bar\varepsilon$, $\bar\tau_0\bar\varepsilon \le \frac14$ and $(\pi^2 + 4\bar\tau_0^2)\bar\varepsilon \le 9$. Put $P = \pi^2 + 4\tau_0^2$ and $\sigma = \tau_0 - \tau$. Then, with the constants of (7.3),*

$$\begin{aligned}
&F(0) \ge (\sigma - \tfrac12\sigma^2)(1 - \tfrac12\bar k\pi^2\varepsilon) - 2k_0\tau_0^2\varepsilon - 2\bar d\bar\tau_0^3\varepsilon^2 - \tfrac{1}{24}\bar\mu(\pi^4 + P^2)\varepsilon^2,\\
&|F'(0)| \le \varepsilon D_1, \qquad D_1 = \bar k\pi(1+2\tau_0) + \bar\mu(\tfrac{\pi^3}{6} + \tfrac{P^2}{24} + \tfrac{P^{3/2}}{6})\varepsilon,\\
&F''(0) \ge e^{-\sigma} - \bar k\varepsilon(1+\tfrac{\pi^2}{2}) - \bar\mu D_2\varepsilon^2, \qquad D_2 = \tfrac{\pi^2}{2} + \tfrac{P^2}{24} + \tfrac{P^{3/2}}{3} + \tfrac{P}{2}.
\end{aligned}$$

Proof. Here $t = \tau\varepsilon \le \tau_0\varepsilon \le \frac14$, and $|x_0| \le |w_0| = \sqrt{(\pi^2+4\tau^2)\varepsilon} \le \sqrt{P\varepsilon} \le 3$. So Lemma 7.2 applies with $R = 3$ and $\mu \le \bar\mu$. Also $x_0^4 = \pi^4\varepsilon^2$, $|w_0|^4 \le P^2\varepsilon^2$, $x_0^3/a = \pi^3\varepsilon^2$, $|w_0|^3/a \le P^{3/2}\varepsilon^2$, $x_0^2\varepsilon = \pi^2\varepsilon^2$ and $|w_0|^2\varepsilon \le P\varepsilon^2$. In (7.6) use $1 - C = 1 - e^{-\sigma} \ge \sigma - \frac12\sigma^2$, $k \le \bar k$, $C \le 1$, $\tau \le \tau_0$ and $k \le k_0 + \bar d t \le k_0 + \bar d\tau_0\varepsilon$. In (7.7) use $|1 - C - 2C\tau| \le 1 + 2\tau_0$. In (7.8) drop the two nonnegative terms $2Ck\tau^2\varepsilon$ and $4Ck\tau\varepsilon$, and use $(1-C) + \frac12 C\pi^2 \le 1 + \frac12\pi^2$. □

7.4 Proof of Theorem 1.4

Let $Q > 0$ and choose $\sigma = \varepsilon s$ with $s = 2k_0\tau_0^2 + Q\varepsilon$, that is,

$$t = \frac{\tau_0 - \sigma}{a^2} = t_m - 2k_0 t_m^2 - Q\varepsilon^3 = t_m - S_2 t_m^2 - Q\,a^{-6},$$

by (7.2). If $t < 0$ there is nothing to prove, since $\Lambda_{c,p} > 0$. Otherwise Lemma 7.3 gives, after dropping the nonnegative term $\frac14\bar k\pi^2\sigma^2\varepsilon$,

$$\frac{F(0)}{\varepsilon^2} \ge Q - \frac{s^2}{2} - \frac{\bar k\pi^2}{2}s - 2\bar d\bar\tau_0^3 - \frac{\bar\mu}{24}(\pi^4 + P^2).$$

Write $F_2 = e^{-\varepsilon s} - \bar k\varepsilon(1+\frac{\pi^2}{2}) - \bar\mu D_2\varepsilon^2$. Then $F(0)F''(0) > F'(0)^2$ follows from $F_2 > 0$ and

$$Q > \frac{s^2}{2} + \frac{\bar k\pi^2}{2}s + 2\bar d\bar\tau_0^3 + \frac{\bar\mu}{24}(\pi^4 + P^2) + \frac{D_1^2}{F_2}. \tag{7.9}$$

Indeed, (7.9) gives $F(0) > \varepsilon^2 D_1^2/F_2 \geq F'(0)^2/F''(0)$. Every term on the right of (7.9) increases with τ_0 and with ε, and F_2 decreases with both. So (7.9) holds for all $\tau_0 \leq \bar\tau_0$ and $\varepsilon \leq \bar\varepsilon$ once it holds at $(\bar\tau_0, \bar\varepsilon)$, with k_0 replaced by an upper bound.

For (i) take $\bar\tau_0 = \log 2$ and $\bar\varepsilon = 4/\log^2 29$, so $\bar\tau_0\bar\varepsilon < 0.2446$ and $(\pi^2 + 4\bar\tau_0^2)\bar\varepsilon < 4.16$. With $Q = \frac{1}{50}$, interval arithmetic gives $F_2 > 0.967$ and the right side of (7.9) below 0.01449. For (ii) take $\bar\tau_0 = 2$ and $\bar\varepsilon = 4/\log^2 293$, so $\bar\tau_0\bar\varepsilon < 0.2480$ and $(\pi^2 + 16)\bar\varepsilon < 3.21$. With $Q = \frac{1}{10}$, interval arithmetic gives $F_2 > 0.977$ and the right side below 0.0680. In both cases $\mathscr{L}_0(H_t^{c,p}; \pi/a) < 0$, and Proposition 5.1 gives $\Lambda_{c,p} > t$.

For fixed c and $p \to \infty$, $\varepsilon \to 0$. The hypotheses of Lemma 7.3 then hold with $\bar\tau_0 = \tau_0$, and the right side of (7.9) tends to a finite limit. Any larger Q works for all large p, which gives the $O(t_m^3)$ statement, since $a^{-6} = t_m^3/\tau_0^3$. □

Corollary 7.4 (under RH)**.** *Assume RH. For* $\frac{1}{2} \leq c < 1$ *and* $p \geq 29$,

$$t_m - S_2 t_m^2 - \tfrac{1}{50} a^{-6} < \Lambda_{c,p} \leq \frac{\delta^2}{2} = t_m\, g(c), \qquad g(c) = \frac{\operatorname{arccosh}^2(1/c)}{2\log(1/c)} = 1 + \frac{1-c}{3} + O((1-c)^2).$$

Proof. Combine Theorem 1.4(i) with Proposition 3.3(a), and note $\delta^2/2 = \operatorname{arccosh}^2(1/c)/(2a^2)$. The expansion of g follows from $\operatorname{arccosh}(1+\eta)^2 = 2\eta - \frac{1}{3}\eta^2 + O(\eta^3)$ and $\log(1+\eta) = \eta - \frac{1}{2}\eta^2 + O(\eta^3)$ with $1/c = 1 + \eta$. □

8 Numerical collision times

This section is numerical evidence, not a certificate. We followed the k-th off-line zero of $H_t^{c,p}$ from its position $(2k+1)\pi/a + i\delta$ at $t = 0$ for $k = 0, 1, 2$, with adaptive steps in t. Near the axis we solved $H = H' = 0$ by Newton's method in (t, x). The integrals were computed with the trapezoid rule on the contour $\operatorname{Im} u = \frac{1}{4}$, which was stable to about thirteen digits under grid refinement, and Theorem 1.2 was checked against direct quadrature to relative accuracy 10^{-13}. Let $T_0(c,p)$ denote the computed collision time of the lowest pair ($k = 0$).

p	c	t_m	$T_0(c,p)$	$(t_m - T_0)/(S_2 t_m^2)$	Thm. 1.4	$\delta^2/2$
2	0.5	5.7708	5.04322	0.946	–	7.220
3	0.5	2.2972	2.17782	0.979	–	2.874
5	0.5	1.0704	1.04416	0.990	–	1.339
11	0.5	0.48220	0.476849	0.9957	–	0.6033
29	0.5	0.244525	0.243146	0.998	0.242265	0.3059
41	0.5	0.201049	0.200117	0.9982	0.199627	0.2515
43	0.5	0.195989	0.195103	0.9983	0.194650	0.2452
47	0.5	0.187038	0.186231	0.998	0.185837	0.2340
101	0.5	0.130173	0.129781	0.99884	0.129649	0.1629
1009	0.5	0.057954	0.0578767	0.99948	0.0578650	0.0725
10007	0.5	0.032679	0.0326543	0.99971	0.0326522	0.0409
2	0.9	0.87718	0.857882	1.085	–	0.908
101	0.9	0.019787	0.0197776	1.0017	0.0196451	0.0205
10007	0.9	0.0049673	0.00496673	1.0004	0.0049646	0.00514

Table 2: Computed lowest-pair collision times (floating point) and the rigorous lower bound of Theorem 1.4, where it applies. The last column is the upper bound for $\Lambda_{c,p}$ under RH.

The computations show the following.

(1) In the twelve cases in which three pairs were tracked (all rows except $p = 29$ and $p = 47$), the lowest pair collides last among them. The higher pairs do not collide in order of k: for $p = 3$ the pair near $x = 28.6$, next to the first zero of ξ, collides first. Pairs with $k \geq 3$ were not tracked.

(2) For $c = \frac{1}{2}$ we also computed $T_0(\frac{1}{2}, p)$ for every prime $p \leq 47$; Table 2 lists some of these values. They decrease in p and exceed 0.2 exactly for $p \leq 41$, with $T_0(\frac{1}{2}, 43) \approx 0.19510$ and $T_0(\frac{1}{2}, 47) \approx 0.18623$.

(3) The ratio $(t_m - T_0)/(S_2 t_m^2)$ tends to 1, consistent with $T_0 = t_m - S_2 t_m^2 + O(t_m^3)$. The next correction has opposite signs for $c = \frac{1}{2}$ and $c = 0.9$.

(4) The collision height x^* approaches $2\pi/\log p = \pi/a$; for $(p, c) = (10007, \frac{1}{2})$, $x^* = 0.6816$ against $\pi/a = 0.68214$.

Remark 8.1 (formal upper half). The derivation of Section 7 also predicts the time at which the lowest pair actually collides. By (7.6)–(7.8), the two equations $F(\theta) = F'(\theta) = 0$ have, formally, a solution with $\theta = O(\varepsilon)$ and $1 - C = 2k_0\tau_0^2\varepsilon + O(\varepsilon^2)$. Since $1 - C = 1 - e^{\tau - \tau_0}$, this is $T_0 = t_m - S_2 t_m^2 + O(t_m^3)$, in agreement with Table 2. Turning this into an upper bound for $\Lambda_{c,p}$ would require showing that every other zero of $H_t^{c,p}$, including those inherited from H_0, is real by time T_0. We do not attempt this.

9 What upper-bound proofs must use

Corollary 9.1. *Let $\mathcal{P}$ be a collection of properties of real even kernels, and suppose every property in $\mathcal{P}$ holds for $\Phi_{1/2,p}$ for some prime $p \leq 41$. Then $\mathcal{P}$ does not imply $\Lambda_K \leq 0.2$. If the properties hold for $\Phi_{1/2,43}$ or $\Phi_{1/2,47}$, then $\mathcal{P}$ does not imply $\Lambda_K \leq 0.1787854$.*

Proof. Otherwise Theorem 1.3 would be contradicted with $K = \Phi_{1/2,p}$. □

The content of the corollary is the list of properties that the controls share with Φ. For $0 < c < 1$ and every prime p, Sections 2 and 3 show:

(S1) $\Phi_{c,p}$ is positive and even, and decays like $\exp(-\pi e^{4|u|-4a})$.

(S2) $H_0^{c,p}$ has exactly the real zeros of H_0, with the same multiplicities, and $1-c \leq H_0^{c,p}/H_0 \leq 1+c$ on $\mathbb{R}$.

(S3) The Dirichlet coefficients are positive and multiplicative, and the Euler factor at every prime $q \neq p$ is that of ζ; equivalently, every Hecke relation at $q \neq p$ holds (Remark 2.3).

(S4) The Dirichlet series lies in $\mathcal{S}^\sharp$ with the gamma factor $\Gamma(s/2)$ of ζ and conductor p^2, and $\Lambda_{c,p} \geq 0$, as for ζ.

(S5) The heat flow is determined by the heat flow of Φ (Theorem 1.2).

They do not share:

(D1) the Euler factor of ζ at p, equivalently conductor one, or the full level-one modularity of the theta function;

(D2) real-rootedness of H_0: $H_0^{c,p}$ has the non-real zeros $\pm(2k+1)\pi/a \pm i\delta$ for every $k \geq 0$, so at every height; the lowest have real part $\pi/a = 2\pi/\log p \leq 9.1$;

(D3) the values of ξ away from the zeros of $\cos ax$.

The existing proofs of $\Lambda \le 0.22$ and $\Lambda \le 0.2$ [11, 9] combine three ingredients. The first is a numerical verification of RH up to a large height, which is an input of type (D2) and (D3). The second is a barrier computation by direct evaluation of the genuine heat-flowed function, again (D3). The third is an effective approximation at large height built from the Dirichlet coefficients of ζ, with Euler mollifiers at small primes, which is (D1). The controls fail the first input already at height $\pi/a \le 9.1$, where their lowest non-real pair lies. So these proofs are consistent with Corollary 9.1. The corollary says something about arguments that avoid such inputs.

A structural argument for $\Lambda \le 0.2$ uses no direct evaluation of ξ. Examples are identities, positivity statements or symmetries of the theta function and its lifts. Such an argument must use, for each prime $p \le 41$, an input that distinguishes ζ from $\Phi_{1/2,p}$, such as the Euler factor of ζ at p or level-one modularity. Finitely many Euler factors or Hecke relations cannot suffice for $\Lambda \le 0$ on their own: for every finite set S of primes and every $p \notin S$, the controls $\Phi_{c,p}$ with $c < 1$ share the factors at S and have $\Lambda_{c,p} > 0$. Hypotheses beyond those shared by the controls, for instance quantitative information about ξ, are not affected by this observation.

Remark 9.2. For $c \ge 1$ the controls are RH-equivalent twins of Φ (Theorem 1.1(c)). Under RH, $\Lambda_{c,p} = 0$ for $c \ge 1$ and $\Lambda_{c,p} > 0$ for $0 < c < 1$, so the transition between $\Lambda_{c,p} > 0$ and $\Lambda_{c,p} \le 0$ in the family is governed by the temperedness of one local factor. This is the sense in which the screen of Corollary 9.1 is arithmetic and not analytic.

10 Open problems

(1) Prove $\Lambda_{c,p} \le t_m - S_2 t_m^2 + O(t_m^3)$, at least under RH, by showing that all other zeros of $H_t^{c,p}$ are real by the time the lowest pair collides.

(2) The witnesses of Theorem 1.3 lie at height below 10, near the lowest off-line pair, which numerically collides last. Construct controls whose last collision occurs at large height, for example from Hecke polynomials at several primes, so as to screen arguments that use only information at large height.

(3) Extend the construction to Dirichlet L-functions and to other members of $\mathcal{S}^\sharp$, where Dobner's theorem gives $\Lambda \ge 0$ throughout.

(4) Determine the minimal arithmetic input. Does any upper bound below $\max_{p \le 41} \Lambda_{1/2,p}$ follow from the Euler factors at the primes up to 41 together with properties (S1)–(S5)?

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This manuscript was drafted for Edward Baker with substantial assistance from Claude Opus 5.5 (Anthropic; configured identifier `claude-opus-5-5`), working in Claude Cowork at reasoning effort max. The assistant contributed the mathematical derivations, the computer programs, the drafting and the internal checks. The Euler-factor controls were introduced in an internal review of the investigation portfolio in the repository `ebbaker/shifted-zeta-positivity`, and their first study is recorded in `papers/investigations-2/active/19_euler_factor_controls`. An internal review there by GPT-6 Astra (Codex) corrected two points of that first study, and those corrections are used here. Tracked collision times are numerical estimates of the last collision of a branch, not values of $\Lambda_{c,p}$. Near an isolated collision the raw witness margin shrinks, as in Remark 5.3.

The certificates of Theorem 1.3 and Lemma 7.1, and the inequality (7.9) in Theorem 1.4, are produced by a standard-library program in the `numerics` subfolder of that folder, which also writes a JSON record. The floating-point data of Section 8 come from a second program in the same subfolder:

```
check_euler_control_witnesses_claude_20261014.py
EULER_CONTROL_WITNESS_CERTIFICATES_20261014.json
check_euler_factor_controls_claude_20261013.py
```

An internal review by a separate Claude instance recomputed every witness and moment with independent high-precision quadrature. It found that the first version of the certificate program rounded negation in Python's default 28-digit context, so that its enclosures were not rigorous, although the computed values were correct to at least 24 significant digits. The program was corrected as described in Section 6 and the certificates were regenerated; the new enclosures contain the independent 75-digit values. These checks, and the parallel internal reviews, are not independent specialist validation. Independent review of the arguments and the code is still needed. No claim of literature priority is made.

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